

Domination–packing ratio for planar and unit disk graphs

W. Cames van Batenburg*

July 15, 2026

Abstract

The *domination number* $\gamma(G)$ of a graph G is the smallest possible size of a vertex set that intersects every radius-1 ball of G , and the *packing number* $\rho(G)$ is the maximum number of pairwise vertex-disjoint radius-1 balls. We prove that $\frac{\gamma(G)}{\rho(G)} \leq 5$ for every planar graph, and $\frac{\gamma(G)}{\rho(G)} \leq \frac{18\sqrt{3}}{\pi} \approx 9.924$ for every unit disk graph, thus yielding Erdős–Pósa-type bounds for the hypergraph of radius-1 balls in the two graph classes. This improves upon results of Gutiérrez and Paul, and Dúcz and Gujgiczer, who in turn lowered bounds of Bonamy, Csikós, Gujgiczer and Yuditsky, and Böhme and Mohar. For both graph classes, the best known lower bound on the optimal constant remains 3.

1 Introduction

Duality between packing and covering is a well-studied theme in structural graph theory: given a family of ‘nice’ objects, one seeks either many pairwise disjoint nice objects, or a small set that intersects every nice object. Statements of this type fall under the umbrella of *Erdős–Pósa* theorems, which can often be encoded in terms of some class $\mathcal{H}$ of hypergraphs. For every hypergraph $H \in \mathcal{H}$, the nice objects correspond to the hyperedges of H . If ρ denotes the maximum number of pairwise disjoint hyperedges and γ the minimum size of a vertex set that intersects every hyperedge of H , then a *bounding function* for $\mathcal{H}$ (if it exists!) is a function f such that $\gamma \leq f(\rho)$ for every $H \in \mathcal{H}$. In the foundational result of Erdős and Pósa [EP65], the hyperedges are the cycles of some fixed graph: they proved that every graph either contains k vertex-disjoint cycles, or a set of $\mathcal{O}(k \log k)$ vertices meeting every cycle. Thus, for cycles there exists a bounding function $f(\rho) = \mathcal{O}(\rho \log \rho)$. Since then, the same question has been studied for many more structured objects, such as directed cycles [RRST96], matroid circuits [GK09], cycles in group-labelled graphs [GHK⁺25] and graph minor models [RS86, CHJR19], see also the survey [RT17].

A particularly relevant special case for this note is that of *ball hypergraphs*. Given a graph G , a ball hypergraph of G has vertex set $V(G)$ and hyperedges given by graph balls $B_r(v) = \{u \in V(G) : \text{dist}_G(u, v) \leq r\}$, possibly with different radii. In [BCE⁺21], Bousquet et al. proved that for every graph Γ there exists a constant c such that every ball hypergraph of a Γ -minor-free graph satisfies $\gamma \leq c \cdot \rho$. In particular this gave a linear bounding function for ball hypergraphs of planar graphs, even when not all balls have the same radii, confirming a conjecture of Chepoi, Estellon and Vaxès [CEV07]. Earlier progress, with constants depending on the radius or on weak colouring numbers, was obtained by Dvořák [Dvo13, Dvo19].

*Département d’Informatique, Université libre de Bruxelles, Belgium (w.p.s.camesvanbatenburg@gmail.com). Supported by the Belgian National Fund for Scientific Research (FNRS).

In this note we focus on the very special case that all balls have radius 1. Then the covering problem reduces to the usual domination problem. More precisely, from now on, for a graph G , the *domination number* $\gamma(G)$ is the minimum size of a set $D \subseteq V(G)$ such that its closed neighbourhood $N_G[D]$ is the whole vertex set $V(G)$, equivalently, such that D has non-empty intersection with every radius-1 ball. And the *packing number* $\rho(G)$ is the maximum size of a set $P \subseteq V(G)$ whose closed neighborhoods are pairwise disjoint, equivalently, whose vertices are pairwise at distance larger than 2. Because the term ‘packing number’ is heavily overloaded in graph theory, referring to numerous very different notions of disjoint structures, a common alternative name for $\rho(G)$ is the *2-independence number*.

Clearly $\rho(G) \leq \gamma(G)$. This inequality is sharp for trees [MM75] and strongly chordal graphs [Far84], but it can fail arbitrarily badly even in sparse graphs: Dvořák [Dvo13] constructed bipartite 3-degenerate graphs with 2-independence number 2 and unbounded domination number. An elegant basic positive bound is $\gamma(G) \leq \Delta(G)\rho(G)$ [HLR11], where $\Delta(G)$ denotes the maximum degree of G . Studies of the case $\Delta(G) = 3$ and other variants can be found in [GG25, CK24, BGH26]. For line graphs there is a matching interpretation: $\gamma(L(H))$ is the minimum size of a maximal matching of H , whereas $\rho(L(H))$ is the maximum size of an induced matching; see [Cam22] for related conjectures and context in the $\Delta(H) = 3$ setting. Recently, Bonamy, Dvořák, Michel and Mikšaník [BDMM26] gave an exact characterization of the monotone graph classes of bounded average degree that admit a linear bounding function $\gamma(G) \leq c \cdot \rho(G)$, for some constant c that only depends on the class.

In a recent work, Bonamy, Csikós, Gujgiczer and Yuditsky [BCGY25] undertook a systematic study of graph classes with constant domination-packing ratio. Among other results, they proved that $\gamma(G) \leq 10\rho(G)$ for planar graphs, thus lowering a $\gamma(G) \leq 20\rho(G) - 9$ bound of Böhme and Mohar [BM03]. They also showed that $\gamma(G) \leq 32\rho(G)$ for every *unit disk graph* G , which by definition is a graph admitting a representation $p : V(G) \rightarrow \mathbb{R}^2$ such that, for distinct vertices u, v , we have $uv \in E(G)$ if and only if the Euclidean distance between $p(u)$ and $p(v)$ is ≤ 1 .

In two separate works, the bounds for both classes have since been improved. Dúcz and Gujgiczer [DG26] improved the planar bound from 10 to 7, while the best known lower bound for planar graphs is 3 due to a graph on nine vertices exhibited by MacGillivray and Seyfarth [MS96]. For unit disk graphs, Gutiérrez and Paul [GP26] improved the bound from 32 to 16 and also gave an infinite family with ratio 3. In this note we further lower the upper bounds for both graph classes.

Theorem 1.1. *For every finite simple planar graph G ,*

$$\gamma(G) \leq 5\rho(G).$$

Similar to the arguments in [DG26, BCGY25], the proof proceeds via a stronger theorem in terms of refinements $\gamma_X(G)$ and $\rho_X(G)$ of $\gamma(G)$ and $\rho(G)$. These keep track of domination and packing on an arbitrary subset $X \subseteq V(G)$, which helps overcome the technical difficulty that a packing on a subgraph may no longer be a packing in the full graph because the addition of a vertex (plus its incident edges) may decrease distances. We proceed by minimum counterexample and use a discharging argument. The main reason why we were able to improve on the argument of [DG26] is the simple yet effective Lemma 2.4.

Theorem 1.2. *For every finite unit disk graph G ,*

$$\gamma(G) \leq \frac{18\sqrt{3}}{\pi} \rho(G) < 10\rho(G).$$

The proof takes a maximal independent (and hence dominating) set I of the unit disk graph, considers the points $X \subseteq \mathbb{R}^2$ that represent I , and then uses a randomly shifted triangular grid to extract a large subset $Y \subseteq X$ of points that are at pairwise Euclidean distance > 2 . The vertices corresponding to Y have disjoint closed neighborhoods and thus form a packing.

2 Planar graphs

In this section we prove Theorem 1.1. The main difficulty in an inductive domination–packing argument is that deleting a vertex may change distances between the remaining vertices. We use the following refinements to keep track of the vertices that are relevant on both sides of the inequality.

Definition 2.1. Let G be a graph and let $X \subseteq V(G)$.

- (i) A set $D \subseteq V(G)$ is an X -dominating set if $X \subseteq N_G[D]$. The minimum cardinality of an X -dominating set is denoted by $\gamma_X(G)$.
- (ii) A set $P \subseteq X$ is an X -packing if $\text{dist}_G(p, q) \geq 3$ for all distinct $p, q \in P$. Equivalently, the closed neighborhoods of distinct vertices of P are disjoint. The maximum cardinality of an X -packing is denoted by $\rho_X(G)$.

Writing $\bar{X} = V(G) \setminus X$ for the complement of a set X in $V(G)$, we note that Definition 2.1 is effectively the same as definitions of Dúcz and Gujgiczer [DG26], who adapted a definition introduced by Bonamy, Csikós, Gujgiczer and Yuditsky [BCGY25]. However, what we call an X -dominating set would be an $\bar{X}$ -dominating set in their terminology. We have made this notational swap, because we found it more intuitive to work with an X -dominating set that is in fact a set that dominates X , and that an X -packing is a packing whose vertices must be chosen from X .

With the choice $X = V(G)$, Definition 2.1 yields $\gamma_{V(G)}(G) = \gamma(G)$ and $\rho_{V(G)}(G) = \rho(G)$. Consequently, Theorem 1.1 will be a direct corollary of the following stronger statement.

Theorem 2.2. For every finite simple planar graph G and every $X \subseteq V(G)$,

$$\gamma_X(G) \leq 5\rho_X(G).$$

2.1 A minimum counterexample

Suppose for a contradiction that Theorem 2.2 is false, and choose a counterexample (G, X) minimizing $|X|$, and subject to that minimizing $|V(G)| + |E(G)|$. Here $X \subseteq V(G)$ is the set that must be dominated and from which the packing vertices must be chosen. If G is disconnected, $G_1, \dots, G_t$ are the components of G and $X_i = X \cap V(G_i)$, then $\gamma_X(G) = \sum_{i=1}^t \gamma_{X_i}(G_i) \leq 5 \sum_{i=1}^t \rho_{X_i}(G_i) = 5\rho_X(G)$, so we may assume that G is connected. It is easy to see that a graph with at most one vertex is not a counterexample, so $|V(G)| \geq 2$. The reductions in Lemma 2.3 are also deduced in [DG26], but we include the proof for completeness.

Lemma 2.3. The following statements hold.

- (a) $\bar{X}$ is independent.
- (b) Every vertex of X has degree at least 6.
- (c) Every vertex of $\bar{X}$ has degree at least 2.

Proof. For (a), suppose that $y_1 y_2 \in E(G)$ with $y_1, y_2 \in \bar{X}$, and let $H = G - y_1 y_2$. By minimality, there are an X -dominating set D' and an X -packing P' in H such that $|D'| \leq 5|P'|$. Restoring

the edge y_1y_2 (or adding any edge, for that matter) cannot spoil X -domination. Thus D' is also an X -dominating set in G . Restoring y_1y_2 cannot spoil the packing property either. Indeed, a hypothetical new path of length ≤ 2 must traverse the new edge y_1y_2 and hence has at least one endpoint in $\{y_1, y_2\}$. Since $P' \subseteq X$ and $y_1, y_2 \notin X$, no such path has both endpoints in P' . Thus P' is also an X -packing in G , a contradiction.

For (b), let $v \in X$ and suppose that $d_G(v) \leq 5$. Since G is connected and has at least two vertices, $d_G(v) \geq 1$. Set

$$X' = \{u \in X \setminus \{v\} : \text{dist}_G(u, v) \geq 3\}.$$

By minimality of $|X|$, there are an X' -dominating set D' and an X' -packing P' in G satisfying $|D'| \leq 5|P'|$. Define

$$D = D' \cup N_G(v) \quad \text{and} \quad P = P' \cup \{v\}.$$

We first verify domination. Every vertex of X' is dominated by D' in G . Every vertex of $X \setminus X'$ is either v itself or is at distance at most two from v in G , and is therefore dominated by $N_G(v)$. Here v is dominated because $d_G(v) \geq 1$. Hence D is an X -dominating set of G .

Next, $P \subseteq X$. Since $P' \subseteq X'$, every vertex of P' is at distance at least three from v in G . Since P' is an X' -packing in G , it follows that P is an X -packing of G . Finally, $|D| \leq |D'| + d_G(v) \leq 5|P'| + 5 = 5|P|$, a contradiction.

For (c), let $y \in \bar{X}$ have degree at most one, and put $H = G - y$. Since $y \notin X$, the set X is contained in $V(H)$. By minimality, there are an X -dominating set D' and an X -packing P' in H such that $|D'| \leq 5|P'|$. The set D' remains an X -dominating set after y is restored. Restoring an isolated vertex or a leaf cannot shorten the distance between two vertices already present in H , so P' remains an X -packing. This contradicts the choice of (G, X) . $\square$

The next lemma is the key local reducibility statement. In particular it implies that after removal of a vertex $y \in \bar{X}$, at least two of its neighbours must be at distance ≥ 3 .

Lemma 2.4. *For every $y \in \bar{X}$ and every $a \in N_G(y)$, there exists $c \in N_G(y) \setminus \{a\}$ such that*

$$\text{dist}_{G-y}(a, c) \geq 3.$$

Proof. By Lemma 2.3(c), y has at least two neighbors. Suppose for a contradiction that some $a \in N_G(y)$ satisfies $\text{dist}_{G-y}(a, c) \leq 2$ for every $c \in N_G(y) \setminus \{a\}$. Let $H = G - ya$. By minimality, there are an X -dominating set D' and an X -packing P' in H with $|D'| \leq 5|P'|$. As X -domination is preserved under edge-addition, D' is also an X -dominating set in G . Suppose that restoring edge ya spoils the packing property. Since $P' \subseteq X$ and $y \in \bar{X}$, we have $y \notin P'$. Therefore the only possible new path of length at most two with both endpoints in P' is a path ayc , where $c \in N_G(y) \setminus \{a\}$ and $a, c \in P'$. By assumption, however, a and c are already at distance at most two in $G - y$, which is a subgraph of H . This contradicts that P' is a packing in H . Hence P' is also an X -packing in G , a contradiction. $\square$

2.2 Discharging

Fix a plane embedding of G . The length $\ell(f)$ of a face f is the length of its facial boundary walk, with repeated vertices and edges counted with multiplicity.

Definition 2.5 (Corner and $\bar{X}$ -corner). Write the cyclic boundary walk of a face f as

$$(v_0, e_1, v_1, \dots, e_{\ell(f)}, v_{\ell(f)} = v_0).$$

For $0 \leq i < \ell(f)$, the vertex v_i in this cyclic walk together with the angular sector of f between the preceding and following edges, is a *corner of f at v_i* . It is an $\bar{X}$ -corner if $v_i \in \bar{X}$. Note that a

cutvertex may determine several corners of the same face. Let $k(f)$ be the number of $\bar{X}$ -corners of f , counted with multiplicity.

While in general cut vertices may prevent a face from being bounded by a cycle, we do not have that problem for faces of length three or four. Recalling that G is simple loopless by assumption and has minimum degree at least two by Lemma 2.3, the next lemma is immediate.

Lemma 2.6. *Every face has length ≥ 3 , and every face of length $\in \{3, 4\}$ is bounded by a cycle.*

Since $\bar{X}$ is independent, no two consecutive corners of a facial boundary walk are $\bar{X}$ -corners. Therefore

$$k(f) \leq \left\lfloor \frac{\ell(f)}{2} \right\rfloor. \quad (1)$$

Now we start the discharging argument. We give every vertex v and face f initial charge

$$\mu(v) = d_G(v) - 6 \quad \text{and} \quad \mu(f) = 2\ell(f) - 6.$$

Euler's formula, together with $\sum_v d_G(v) = \sum_f \ell(f) = 2|E(G)|$ then gives that the total sum of the charges $\sum_{v \in V(G)} \mu(v) + \sum_f \mu(f) = (2|E| - 6|V|) + (4|E| - 6|F|) = -12$ is negative. As is the standard approach, we will redistribute the charges so that in the end each vertex and face will have nonnegative charge, while also always keeping the total sum of the charges the same -12 , thus arriving at a contradiction. For that redistribution, we will need only one rule.

Discharging rule. *Every face f with $\ell(f) \geq 4$ and $k(f) > 0$ distributes its entire charge equally among its $\bar{X}$ -corners: each such corner receives $\mu(f)/k(f)$. The charge received by a corner is subsequently passed to the vertex determining that corner.*

Note that a face with $k(f) = 0$ keeps its charge, and triangular faces have charge zero. Every $\bar{X}$ -corner of a face f receives charge $(2\ell(f) - 6)/k(f)$, which by (1) is at least

$$1 \quad \text{if } \ell(f) = 4, \quad 2 \quad \text{if } \ell(f) \geq 5. \quad (2)$$

Indeed, a face of length four has charge 2 and at most two $\bar{X}$ -corners, while for $\ell(f) \geq 5$ we have $2\ell(f) - 6 \geq 2\lfloor \ell(f)/2 \rfloor \geq 2k(f)$. Every face therefore finishes with nonnegative charge. Every vertex of X also finishes with nonnegative charge by Lemma 2.3(b), as does every vertex of $\bar{X}$ of degree at least six.

Let $y \in \bar{X}$ have degree $d \in \{2, 3, 4, 5\}$. List its neighbors cyclically as $v_1, \dots, v_d$, and let f_i be the face in the corner between yv_i and yv_{i+1} , with indices modulo d . Define

$$w_i = \begin{cases} 0, & \text{if } \ell(f_i) = 3, \\ 1, & \text{if } \ell(f_i) = 4, \\ 2, & \text{if } \ell(f_i) \geq 5. \end{cases}$$

By (2), the charge received by y is at least $\sum_{i=1}^d w_i$. If $w_i = 0$, then Lemma 2.6 implies $v_i v_{i+1} \in E(G - y)$, so then we say that the corner between yv_i and yv_{i+1} is *triangular*. If $w_i = 1$, the same lemma implies there is a path of length two between v_i and v_{i+1} in $G - y$, so then we say the corner between yv_i and yv_{i+1} is *quadrilateral*.

The next and last lemma produces a lower bound on the total charge that a degree ≤ 5 vertex of $\bar{X}$ receives.

Lemma 2.7. For every $y \in \bar{X}$ of degree $d \leq 5$,

$$\sum_{i=1}^d w_i \geq 6 - d.$$

Proof. Suppose for a contradiction that $\sum_{i=1}^d w_i \leq 5 - d$. Then for every case below, we derive that all pairs of neighbors of y are at distance at most two in $G - y$, contradicting Lemma 2.4.

If $d = 5$, then every w_i is zero, so every corner is triangular. Thus the neighbors of y contain the cycle $v_1 v_2 v_3 v_4 v_5 v_1$ in $G - y$, and every two of them are at distance at most two.

If $d = 4$, then all corners are triangular except possibly one quadrilateral corner. Relabeling if necessary, we may assume that the corner between $y v_4$ and $y v_1$ is the only corner that might be quadrilateral. The vertices v_1, v_4 are at distance at most two, and $v_1 v_2 v_3 v_4$ must be a path. Hence every two neighbors of y are at distance at most two in $G - y$.

If $d = 3$, then either every $w_i \leq 1$, in which case each of the three neighbor-pairs is at distance at most two, or one weight is two and the other two are zero. In the latter case the two triangular edges form a path of length two between the remaining pair. Again every two neighbors are at distance at most two in $G - y$.

If $d = 2$, then at least one of the two weights must be ≤ 1 . The corner corresponding to that weight is triangular or quadrilateral, so there is a path of length at most two between the two neighbors of y in $G - y$. $\square$

Proof of Theorem 2.2. A vertex $y \in \bar{X}$ of degree $d \leq 5$ starts with charge $d - 6$ and, by Lemma 2.7, receives at least $6 - d$. Hence it finishes with nonnegative charge. As observed above, every other vertex and every face also finishes with nonnegative charge. This contradicts the total charge -12 . Therefore a minimum counterexample does not exist, and $\gamma_X(G) \leq 5\rho_X(G)$ holds for every finite simple planar graph G and every $X \subseteq V(G)$. $\square$

Proof of Theorem 1.1. Apply Theorem 2.2 with $X = V(G)$. $\square$

3 Unit disk graphs

In this section we prove Theorem 1.2. A subset $P \subset \mathbb{R}^2$ is called *c-separated* if $d(x, y) > c$ for all distinct $x, y \in P$. Here $d(x, y)$ denotes the Euclidean distance between x and y . To prove Theorem 1.2, we first derive an auxiliary lemma that finds a large 2-separated set inside a given 1-separated set. The averaging step in the argument is standard and can be viewed as an instance of Blichfeldt's principle in the geometry of numbers [Cas97]. Related random shift constructions for unit disk graphs appear in [CFKP07] and [EJM23].

Lemma 3.1. Let $X \subset \mathbb{R}^2$ be a finite 1-separated point set. Then there exists a 2-separated subset $Y \subseteq X$ of size $|Y| \geq \frac{\pi}{18\sqrt{3}} |X|$.

Proof. Let Λ be the triangular lattice with shortest nonzero vector of length 3. For instance, take $\Lambda = \left\{ m(3, 0) + n \left(\frac{3}{2}, \frac{3\sqrt{3}}{2} \right) : m, n \in \mathbb{Z} \right\}$. A fundamental parallelogram of Λ has area $\det(\Lambda) = \frac{9\sqrt{3}}{2}$. Let

$$S = \bigcup_{\lambda \in \Lambda} B(\lambda, 1/2),$$

where $B(\lambda, 1/2)$ denotes the open Euclidean disk of radius $1/2$ centred at λ . For a translation vector $t \in \mathbb{R}^2$, define

$$Y_t = X \cap (S + t).$$

We first observe that Y_t is always 2-separated. Indeed, suppose $x, x' \in Y_t$ are distinct. If x and x' lie in the same open disk of radius $1/2$, then $d(x, x') < 1$, contradicting the hypothesis on X . Hence they lie in two distinct translated disks. The centres of distinct disks in $S + t$ are at distance at least 3. Since the disks are open and have radius $1/2$, we get $d(x, x') > 3 - \frac{1}{2} - \frac{1}{2} = 2$. Thus Y_t is 2-separated.

Now choose t uniformly at random from a fundamental parallelogram of Λ . For every fixed point $x \in X$, the probability that $x \in S + t$ is exactly the density of S , namely

$$\frac{\text{area}(B(0, 1/2))}{\det(\Lambda)} = \frac{\pi/4}{9\sqrt{3}/2} = \frac{\pi}{18\sqrt{3}}.$$

Therefore the expected number of points in Y_t is $\mathbb{E}|Y_t| = \sum_{x \in X} \mathbb{P}(x \in S + t) = \frac{\pi}{18\sqrt{3}} |X|$. So for at least one translation t , we have $|Y_t| \geq \frac{\pi}{18\sqrt{3}} |X|$. $\square$

Proof of Theorem 1.2. Fix a unit disk representation $p : V(G) \rightarrow \mathbb{R}^2$ such that, for distinct vertices $u, v \in V(G)$, we have $uv \in E(G)$ if and only if $d(p(u), p(v)) \leq 1$. Let I be a maximal independent set of G . Due to maximality I is dominating, so $\gamma(G) \leq |I|$. Since I is independent in G , we have $d(p(u), p(v)) > 1$ for all distinct $u, v \in I$. Applying Lemma 3.1 to the point set $p(I)$, we obtain a 2-separated subset $Y \subseteq p(I)$ with $|Y| \geq \frac{\pi}{18\sqrt{3}} |I|$.

Let $P = \{u \in I : p(u) \in Y\} \subseteq I$ be the corresponding set of vertices of G . We claim that P is a packing in G . Indeed, if distinct vertices $u, v \in P$ had a common vertex in their closed neighbourhoods, say $w \in N_G[u] \cap N_G[v]$, then due to the triangle inequality $d(p(u), p(v)) \leq d(p(u), p(w)) + d(p(w), p(v)) \leq 2$, contradicting that Y is 2-separated. Thus P is a packing, and so $\rho(G) \geq |P| \geq \frac{\pi}{18\sqrt{3}} |I| \geq \frac{\pi}{18\sqrt{3}} \gamma(G)$. $\square$

4 Concluding remarks

4.1 Planar graphs

MacGillivray and Seyffarth [MS96] gave an example of a planar graph H with diameter 2 and domination number 3. Later, Goddard and Henning [GH02] showed that H is the unique such graph. In part inspired by the fact that $\rho(H) = 1$ and $\gamma(H) = 3$, Dúcz and Gujiczer [DG26] conjectured that $\gamma(G) \leq 3 \cdot \rho(G)$ is the optimal bound for planar G . We were not yet able to reduce the multiplicative constant below 5, let alone make it 3, but it does appear there is a way to strengthen our Theorem 1.1 to a bound for which the same H is sharp, namely $\gamma(G) \leq 5 \cdot \rho(G) - 2$ for every planar G . Most of the proof would stay the same, but the natural modification $\gamma_X(G) \leq 5 \cdot \rho_X(G) - 2$ of Theorem 2.2 needs to be stated for all G, X that are *non-empty*. This necessitates a separate treatment of the case that $X' = \emptyset$ and $\rho_X(G) = 1$ in the proof of Lemma 2.3(b). This in turn needs a proof that if the packing number $\rho_X(G)$ equals 1, then $\gamma_X(G) \leq 3$. For $X = V(G)$, this follows from a theorem of MacGillivray and Seyffarth [MS96, Theorem 1]. For its extension to arbitrary X , the separation and connector arguments of [GH02], particularly Lemma 13, appear relevant but in need of multiple pages of proof. The latter approach was suggested by ChatGPT 5.6 Sol in the final phase of this project, when we uploaded this draft and asked it to find this specific improvement. Since we prefer to keep the note short and easy to follow, we do not pursue that technical extension here. Upon asking the model to lower the multiplicative constant 5, it made some structural progress towards a potential ratio $9/2$, but was still far from closing any argument.

In the introduction we discussed that for every t , there exists $c_t > 0$ such that $\gamma(G) \leq c_t \cdot \rho(G)$ for every K_t -minor-free graph G , due to Bousquet et al. [BCE⁺21]. Their argument was designed to work for arbitrary ball hypergraphs and consequently the constant c_t obtained there

is quite large, of the order $\exp(t(\log t)^{3/2})$. As planar graphs are K_5 -minor-free, the following is a natural optimization problem.

Question 4.1. *What is the smallest constant $c > 0$ such that $\gamma(G) \leq c \cdot \rho(G)$ for every K_5 -minor-free graph G ?*

The graph H mentioned above shows $3 \leq c$ and a result of Böhme and Mohar [BM03] for graphs excluding $K_{4,4}$ as a minor immediately implies $c \leq 31$. Could we have $c \leq 5$?

4.2 Unit disk graphs

Any improvement of the constant $c = \frac{\pi}{18\sqrt{3}}$ in Lemma 3.1 directly implies an improvement $\gamma(G) \leq \frac{1}{c}\rho(G)$ in Theorem 1.2. We suspect that $c \approx \frac{1}{6}$ is achievable, but that the "hail shot" approach of choosing some fixed lattice independent of P , and then taking a (uniformly) random translation, has almost been optimized. Preliminary computations indicate that small gains are possible by slightly adapting the shape and size of the radius $1/2$ balls around the lattice points, or by taking a random rotation in addition to a random translation. However, these modifications make the proofs much longer, more technical and ugly, in return for only minor improvements. So far, the best ratio that we could heuristically obtain in this way is $\gamma(G) \leq \frac{9\pi}{9\sqrt{3}-4\pi}\rho(G) \approx 9.36\rho(G)$ via randomly rotated and translated regular hexagons. For significant further improvements, a more direct approach tailored to the specific structure of the given point sets P and the Euclidean metric may be necessary.

Tool and computational resource disclosure. The author made use of DuckDuckGo, Google, Overleaf, Python, ChatGPT, Claude, laptop, coffee and human neurons.

References

- [BCE⁺21] NICOLAS BOUSQUET, WOUTER CAMES VAN BATENBURG, LOUIS ESPERET, GWENNAËL JORET, WILLIAM LOCHET, CAROLE MULLER, and FRANÇOIS PIROT (2021). [Packing and covering balls in graphs excluding a minor](#). *Combinatorica* **41**(3), 299–318. [↑1, 7](#)
- [BCGY25] MARTHE BONAMY, MÓNICA CSIKÓS, ANNA GUJGICZER, and YELENA YUDITSKY (2025). [On graph classes with constant domination-packing ratio](#). [↑2, 3](#)
- [BDMM26] MARTHE BONAMY, ZDENĚK DVOŘÁK, LUKAS MICHEL, and DAVID MIKŠANÍK (2026). [Characterization of sparse monotone graph classes with bounded domination-to-2-independence ratio](#). [↑2](#)
- [BGH26] GEOFFREY BOYER, WAYNE GODDARD, and MICHAEL A. HENNING (2026). [On the relationships between domination, isolation, and packing](#). [↑2](#)
- [BM03] THOMAS BÖHME and BOJAN MOHAR (2003). [Domination, packing and excluded minors](#). *The Electronic Journal of Combinatorics* **10**(1), N9. [↑2, 8](#)
- [Cam22] WOUTER CAMES VAN BATENBURG (2022). [Minimum maximal matchings in cubic graphs](#). *The Electronic Journal of Combinatorics* **29**(2), P2.36. [↑2](#)
- [Cas97] J. W. S. CASSELS (1997). [An introduction to the geometry of numbers](#). Classics in Mathematics (Springer-Verlag, Berlin and Heidelberg). [↑6](#)
- [CEV07] VICTOR CHEPOI, BERTRAND ESTELLON, and YANN VAXÈS (2007). [Covering planar graphs with a fixed number of balls](#). *Discrete & Computational Geometry* **37**(2), 237–244. [↑1](#)

- [CFKP07] IOANNIS CARAGIANNIS, ALEKSEI V. FISHKIN, CHRISTOS KAKLAMANIS, and EVI PAPAIOANNOU (2007). [Randomized on-line algorithms and lower bounds for computing large independent sets in disk graphs](#). *Discrete Applied Mathematics* **155**(2), 119–136. [↑6](#)
- [CHJR19] WOUTER CAMES VAN BATENBURG, TONY HUYNH, GWENAËL JORET, and JEAN-FLORENT RAYMOND (2019). [A tight Erdős–Pósa function for planar minors](#). *Advances in Combinatorics* **2019**(2), 33 pp. [↑1](#)
- [CK24] EUN-KYUNG CHO and MINKI KIM (2024). [Independent domination versus packing in subcubic graphs](#). *Discrete Applied Mathematics* **357**, 132–134. [↑2](#)
- [DG26] ÁKOS DÚCZ and ANNA GUJGICZER (2026). [Domination and packing in graphs](#). [↑2, 3, 7](#)
- [Dvo13] ZDENĚK DVOŘÁK (2013). [Constant-factor approximation of the domination number in sparse graphs](#). *European Journal of Combinatorics* **34**(5), 833–840. [↑1, 2](#)
- [Dvo19] ZDENĚK DVOŘÁK (2019). [On distance \$r\$ -dominating and \$2r\$ -independent sets in sparse graphs](#). *Journal of Graph Theory* **91**(2), 162–173. [↑1](#)
- [EJM23] LOUIS ESPERET, SÉBASTIEN JULLIOT, and ARNAUD DE MESMAY (2023). [Distributed coloring and the local structure of unit-disk graphs](#). *Theoretical Computer Science* **944**, 113674. [↑6](#)
- [EP65] PAUL ERDŐS and LAJOS PÓSA (1965). [On independent circuits contained in a graph](#). *Canadian Journal of Mathematics* **17**, 347–352. [↑1](#)
- [Far84] MARTIN FARBER (1984). [Domination, independent domination, and duality in strongly chordal graphs](#). *Discrete Applied Mathematics* **7**(2), 115–130. [↑2](#)
- [GG25] RENZO GÓMEZ and JUAN GUTIÉRREZ (2025). [Domination and packing in graphs](#). *Discrete Mathematics* **348**(5), 114393. [↑2](#)
- [GH02] WAYNE GODDARD and MICHAEL A. HENNING (2002). [Domination in planar graphs with small diameter](#). *Journal of Graph Theory* **40**(1), 1–25. [↑7](#)
- [GHK⁺25] J. PASCAL GOLLIN, KEVIN HENDREY, O-JOUNG KWON, SANG-IL OUM, and YOUNGHO YOO (2025). [A unified Erdős–Pósa theorem for cycles in graphs labelled by multiple abelian groups](#). *Mathematische Annalen* **393**(2), 2507–2559. [↑1](#)
- [GK09] JIM GELEN and KASPER KABELL (2009). [The Erdős–Pósa property for matroid circuits](#). *Journal of Combinatorial Theory, Series B* **99**(2), 407–419. [↑1](#)
- [GP26] JUAN GUTIÉRREZ and KAUSTAV PAUL (2026). [Improved domination–packing bounds in claw-free cubic graphs and unit disk graphs](#). [↑2](#)
- [HLR11] MICHAEL A. HENNING, CHRISTIAN LÖWENSTEIN, and DIETER RAUTENBACH (2011). [Dominating sets, packings, and the maximum degree](#). *Discrete Mathematics* **311**(18–19), 2031–2036. [↑2](#)
- [MM75] AMRAM MEIR and JOHN W. MOON (1975). [Relations between packing and covering numbers of a tree](#). *Pacific Journal of Mathematics* **61**(1), 225–233. [↑2](#)
- [MS96] GARY MACGILLIVRAY and KAREN SEYFFARTH (1996). [Domination numbers of planar graphs](#). *Journal of Graph Theory* **22**(3), 213–229. [↑2, 7](#)
- [RRST96] BRUCE A. REED, NEIL ROBERTSON, PAUL D. SEYMOUR, and ROBIN THOMAS (1996). [Packing directed circuits](#). *Combinatorica* **16**, 535–554. [↑1](#)

- [RS86] NEIL ROBERTSON and PAUL D. SEYMOUR (1986). [Graph minors. V. Excluding a planar graph](#). *Journal of Combinatorial Theory, Series B* **41**(1), 92–114. [↑1](#)
- [RT17] JEAN-FLORENT RAYMOND and DIMITRIOS M. THILIKOS (2017). [Recent techniques and results on the Erdős–Pósa property](#). *Discrete Applied Mathematics* **231**, 25–43. [↑1](#)